



Course MSc Economics and Data Analytics Thesis

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Title: *"The Causal Effect of the 2017 Honduran Election Crisis on Human Trafficking Flows into the United States: Evidence from Synthetic Control"*

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In the last decade, economists have been increasingly applying causal inference techniques to questions of migration and illicit flows. This thesis investigates whether the 2017 Honduran presidential election crisis had a causal impact on the trafficking flows into the United States. I construct a counterfactual for Honduras using the Synthetic Control Method (SCM) which uses a donor pool of Latin American countries and covariates including economics indicators, demographics and institutional strength predictors.

The results indicate a temporary divergence between Honduras and its synthetic control in 2017-2019 which dissipates in 2020. A placebo-in-space test yields non-significant p-values suggesting the effects are not statistically distinguishable from noise. Robustness checks included backfilling of reporting gaps and leave-one-out analysis.

The findings suggest that political instability caused by the 2017 election crisis did not cause any persistent increase in trafficking flows. Policy implications and recommendations found that there needs to be more consistent international reporting before any strong causal claims can be made.

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Introduction

Human trafficking is one of the fastest-growing illicit markets in the world, generating enormous profits for criminal networks while targeting the most vulnerable people. Unlike the drug and arms trades, which have defined, standardised definitions, trafficking does not. Trafficking involves the direct exploitation of people, typically through forced labour and sexual exploitation. Although the real scale and cost of trafficking are widely acknowledged, rigorous quantitative research on its drivers and dynamics remains limited. The existing literature has largely focused on “push-pull” frameworks or small-scale micro evidence, leaving open the question of how political instability and institutional breakdowns shape trafficking flows across borders.

This thesis seeks to address this gap by examining whether the 2017 Honduran presidential election crisis led to an increase in trafficking victims detected in the United States. Honduras provides a compelling case study: it is one of the poorest and most deprived countries in Latin America, which is part of the “Northern Triangle” that has been a massive point of debate on migration and trafficking, and it experienced a highly contested election in 2017 that triggered widespread unrest and state repression. The United States represents a major “pull” destination with high demand for vulnerable labour and a large underground commercial sex market. These conditions make the Honduras- U.S corridor a critical setting to study the link between political shocks and trafficking channels.

In order to estimate causal effects, this study applies the Synthetic Control Method (SCM) (Abadie et al. 2010). SCM is used to construct a counterfactual trajectory for Honduras by optimally weighting a combination of Latin American donor countries that did not experience similar political shocks. By comparing the actual Honduras to that of its synthetic counterpart, the analysis isolates a small potential impact of the 2017 election crisis. Statistical inference was then conducted by using placebo (permutation) tests and robustness checks. These tests include donor cleaning and backfilling to account for anomalies in reporting.

The findings suggest that Honduras exhibited a divergence from its synthetic control after 2017, with a corresponding average increase of approximately 135 trafficking victims per year. However, randomisation inference indicates that these results are not statistically significant, and then the robustness checks reveal that the estimated effect is temporary and sensitive to data irregularities from the donor pool. The study therefore concludes that there is no strong evidence of a persistent causal effect of the 2017 Honduran election crisis on trafficking flows into the United States.

The contribution of this thesis is twofold. Firstly, it uses causal inference method laid out by Abadie et al. (2010) and applies it to the context of human trafficking, where empirical evidence is scarce. The policy relevance of this type of research is important to future policy makers. Secondly, it highlights the challenges of measurement and inference in trafficking research, particularly the reliance on administrative data that may be affected by changes in reporting intensity rather than true values. This

is likely because of the illicit market structure nature of the crime. This provides a foundation for future work and underscores the importance of improving international trafficking data collection and a standardised approach.

Literature review

The field of quantitative human trafficking research is small; there is lots of work and funding going into drugs and arms markets. Akee et al. (2010) is one of the pioneering papers in the area of quantitative analysis of human trafficking. They look at the links ethnic conflicts have on international human trafficking. Akee et al. hypothesis is that people leave their home countries in order to find better economic prospects elsewhere, but along this chain, they are left vulnerable to being trafficked. When there is internal conflict in the host country, it makes the push factors to leave even more extreme. In the research of human trafficking, it is quite common to talk about “push” and “pull” factors. Push factors are reasons as to why people leave their home country in the first place, while pull factors are reasons why people want to go to host countries. Mahmoud and Trebesch (2010) argue that it is the hope for a better life abroad that puts millions of people at risk of exploitation. Mahmoud and Trebesch broke their paper up into two main areas, demand and supply. On the demand side they found that an increase in migration levels drove down the cost of trafficking a potential victim and therefore allowed the criminal network to scale. On the supply side a high number of vulnerable people are willing to work abroad, giving plenty of supply to the traffickers. Vayrynen (2005) argues that humans are a good commodity, as they do not perish easily and can be bought and sold allowing for low barriers to entry into the market.

Theory would suggest that as the rate of supply of potential victims increase so too does the likelihood of a person be trafficked. In fact, Akee et al. (2010) found that ethnic fragmentation increases the likelihood of trafficking but at a decreasing rate. Cho et al. (2012) discuss the impact that prostitution laws have on the likelihood of human trafficking. They found that most victims are women and girls, and they broke down using a binary regression what effect the prostitution laws in the host country has on trafficking. In their study they concluded that if prostitution was legal in the host country than the demand would increase and so would demand for both legal and illegal supply of people. This conclusion was in contrast to Akee et al. (2010) which found no relationship between prostitution laws and trafficking rates. Cho et al. (2012) also found that scale effect dominates the substitution effect. Showing that the scale effect of legalising prostitution led to an expansion of the prostitution market and thus increases the rate of human trafficking.

Mahmoud and Trebesch (2010) found that using TV awareness campaigns in areas of high migration which would be more prone to becoming targets, had a lower level of reported trafficking

observations versus areas that did not run awareness campaigns. Joarder and Miller (2013) discuss how via the use of migration-debt contracts allows for people who do not have the capital to leave their home country can take on debt in order to leave. Then pay this back once they start working in the host country. Since the migration markets are private, that being governments do not actively encourage or support their citizens from leaving for work elsewhere.

Joarder and Miller (2013) make the argument that the middleman will make the decision to traffic or not traffic an individual. That being if they decide that the economic profit is greater to traffic an individual in the host country than they will, this depends on their own moral code/ethics. Joarder and Miller refer to this as a “two-level decision-making process”. The use of these debt contracts is already targeted at the most vulnerable of individuals which raised the risk profile of them being trafficked just by accessing them. In a study published on the illegal Bangladeshi migrants, it is seen that more than half of the migrants avail of these contracts at levels of around 60% (Joarder and Miller, 2013). As the migrant becomes more desperate to leave the country, the more at risk they will be, there are two main reasons for this. Akee et al. (2010) shows a connect between escalating conflict as an increasing push factor. Joarder and Miller (2013) discuss how debt-migration contracts are structured in the following way. They have a timeframe where they do not need to repay anything (usually maximum one year) and then the payments start to come due. The repayments are not that much if they are being paid with a salary from a host country. They are very high if you need to use a wage from the home country. The traffickers and migrants are in an asymmetric information game where the traffickers know that these repayments are coming due, so they demand more money in order to “move the migrant up the list”. So as the time to repayment comes closer, the more desperate the migrant becomes to take any job and are therefore at an increasing risk of being trafficked.

Hernandez and Rudolph (2015) discuss the drivers of human trafficking in Europe. They build of the theory of push and pull factors by those strong institutions which hinders exploitation opportunities in host countries however, this does not reduce the recruitment operations from source countries.

Hernandez and Rudolph also build on the idea that one someone owes debt in the source country is much easier to push them into being trafficked. In my study this can be seen in the case of the Honduras where people are desperate to leave and will be in a harsh enough situation as to take on debt to escape. They broke their victim pool into two groups, the first was individuals who took on debt and the second was those who are deceived from the prospects of a job offer. Amna Malik (2017) argues the idea that globalisation has a negative effect on human trafficking, that it pushes down cost for traffickers allowing for more incentive to enter the market. Jiang and Lagree (2010) found that as trade decreases, human trafficking rate increases.

A closely related SCM has been used by Peri and Yassenov (2019) in the study of migration shocks due to the 1980 Mariel Boatlift study by Card, D. (1990) This paper was a revisit of the classic Miami

shock which used difference-in-differences except, this time a synthetic Miami was constructed by using the weighted combination of U.S. metropolitan areas. The researchers matched Miami pre boatlift wages and employment in order to show that the post 1980 wage trends for low-educated, non-Cuban do not experience a statistically meaningful decline in wages compared to that of the synthetic counterfactual. Two important features of this paper have been used below in order to build a creditable synthetic counterfactual for Honduras, transparent pre-treatment fit and randomisation inference based on a placebo in place benchmark. Peri and Yassenov (2019) provide a framework for building a synthetic control. Examples include picking a donor selection in a way to avoid structurally dissimilar control. There is also support provided in this paper for using a weight vector that concentrates on a single donor which is what occurs in the baseline synthetic for Honduras. The inference strategy used by Peri and Yassenov is post/pre RMSPE with a pointwise year specific test which are utilized in this thesis. Peri–Yassenov provide a reputable approach to building a concrete design choice which has been used throughout this paper.

Agersnap, Jensen and Kleven (2020) identify a large “pull-factor” of immigration with respect to the generous welfare states that Denmark operates in. By contrast, my study examines the “push-factor” shock of the 2017 Honduran political crises. Denmark offers pull side incentive for migrate inflows and in the Honduras case there is a push side political shock. Both of these research designs are based on the credibility of counterfactuals. Agersnap et al. use policy discontinuities within Denmark and an unaffected control group. I use SCM with pre-treatment fit diagnostics, placebo-in-space tests and ratio/pointwise randomisation inference. While in Agersnap et al. it is shown that Denmark’s welfare reform isolates a clear pull mechanism by being able to use quasi-experimental variation, the SCM in this paper focuses on a push shock in Honduras. Together the two papers provide a solution as to what determinants lead to irregular migration and trafficking flows from both ends of the corridor.

The introduction on cross-board controls is not a sufficient way to “turn off” the movement of people between countries. People who are motivated enough will reconfigure how they move. The changes in movement patterns are relevant to trafficking risk along the Honduras – U.S corridor. Avdan et al. 2023 show that physical barriers increase refugee flows rather than reduce mobility. They study this by using global dyad-year design (1990 – 2016) and instrumental variables. It was found that higher risks and sunk cost deter return and push migrants to convert movements into patterns other forms such as asylum claims (Avdan et al. 2023). For those who are Honduran migrants a similar dynamic imply displacement from known land routes and more covert pathways that go through Mexico and the US border. This means that there is an increasing reliance on illicit intermediators. The substitution raises the price of passage as well as the leverage the criminal intermediaries have over their victims when smuggling them. The supply side effect of hard borders can unintentionally increase the vulnerabilities that anti-trafficking policies are aiming to reduce.

The Case for Honduras and the U.S

Firstly, why Honduras? For this thesis the goal was to test the hypothesis of “is it possible to use quantitative methods to uncover a causal link between human trafficking and instability within a country”. The reason for using Honduras is that there was the most amount of data available for without a break in reporting. (UNODC, World Bank, TIP) Honduras also provides a two strong treatment effects with well-timed institutional shock that are plausible to be exogenous (2009 coup; 2017 election crisis)

The 2009 coup as an exogenous shock

Honduras experienced a military coup on the 29th of June 2009 which included the ousting of President Manuel Zelaya. President Zelaya was elected in 2005 and had proposed a non-binding referendum to ask the people whether they wanted to convene an assemble to rewrite the Honduran Constitution. There were fears that Zelaya, wanted to move towards a Hugo Chávez’s authoritarian model, the interim government of Roberto Micheletti was extensively linked to repressive tactics on peaceful protesters which included at least 14 counts of torture linked to post-coup crackdowns (Physicians for Human Rights, 2014).

Sharp timing and widespread institutional instability make it a good choice for a synthetic control. Known effects of migration due to a weakened institution led to the increase violence and economic disruption. It is because of this increased institutional deterioration that provides a plausible causal link between the coup and regional instability which in turn increased trafficking vulnerability.

Honduras saw a significant increase in outward migration in the years following the coup. These migrants include vulnerable populations such as women and unaccompanied minors (Clemens, 2017). According the to the U.S Trafficking in Persons (TIP) reports, Honduras has struggled to meet minimum standards for the elimination of trafficking due to corruption and a weak law enforcement department (U.S. Department of State, 2022, 2023). As well as that, Honduras is part of the Northern Triangle which is a region that has been central to discussion around migration and trafficking. International agencies such as the UN office on Drugs and Crime (UNODC) and World Bank makes Honduras a viable choice for study on causality. However, the victim reports by the UNODC only start in 2005. With the coup in 2009, there is only 4 years of pretreatment data which would not be enough to build a credible synthetic pre trend.

The 2017 election crisis as a second shock

In 2015 the Supreme Court removed a constitutional ban on presidential re-election which allowed Juan Orlando Hernández to run again for the position (Reuters, 2015), this decision stuck worry of a contagion effect among other countries in the Northern Triangle (EU EOM, 2017). The elections of 2017 created “*serious doubts about the fairness of the election*” according to the European Union (EU EOM, 2017). There was delayed and suspicious vote counts because initial vote counts showed the opposition leader, Nasralla leading by over 5 points with a total of 57% of votes counted. Then the electoral system went offline for a number of hours without an explanation and after it came back on, *Juan Orlando Hernández was re-elected with 42.95% vs Nasrallas 41.42%* (EU EOM, 2017).

After the election counts took place, Honduras erupted into mass protests and long-term violence which lasted until 2018. Curfews were brought in and enforced by the military to try and stop the mass protest, however it is estimated that between 22 – 30+ civilians were killed during this time as well as 1,300 arrests and many injuries (Human Rights Watch, 2019). This was the worst crisis the country experienced since the 2009 military coup. It is this political shock which represents a clear, exogenous shock to Honduran political stability and is why it could be a strong treatment event for causal inference. It includes the three major drivers of what increases the vulnerability of people to be pushed into human trafficking. The breakdown in institution, mass unrest from the protest and evidence of human rights violations (Human Rights Watch, 2019). As well as this political instability creating a major push factor, there was also major economic hardship and poverty in the country. Honduras is one of the poorest countries in Latin America with over 60% of the population living in poverty and very high youth unemployment. It also has one of the highest homicide rates in the world peaking at 86 per 100,000 people in 2011 (UNODC). The combination of chronic poverty problems, widespread violence and weak institutions created the perfect grounds for human trafficking networks to form a corridor through central America and into the United States (Avdan et al. 2023).

The United States of America (U.S) provides a major pull incentive as there is better economic opportunity for employment and a much better standard of living, especially compared to Honduras. Even low wage or informal jobs in the U.S pay more than formal work in Honduras. Traffickers prey on the vulnerable by offering false jobs, fake visas and promising legal pathways that do not exist (Vayrynen, 2005). The U.S has a high demand for cheap and un-skilled labour in domestic work, construction and hospitality sector where many victims of labour trafficking are brought in under debt bondage (Majeed et al. 2017). Lastly, the U.S has a large and profitable underground commercial sex economy with high demand combined with a lack of strong anti-trafficking laws has made the U.S, a top destination for sex trafficking victims (Dank et al. 2014). The estimated city-level underground market for the sex industry is between \$40 - \$290 million, implying high demand side pressure which traffickers can exploit (Dank et al. 2014). This is further supported by work by Cho et al. (2012)

which found that demand side factors like prostitution legalisation or prostitution tolerance can increase the trafficking rates. Because of the factors mentioned above, the U.S functions as a powerful pull factor for trafficking during the period of 2017. The U.S economy was strong and growing after rebounding fully from the post-2008 Great Recession (BEA, 2019). This environment intensified the pull factors for migrants and trafficking victims especially from poor unstable countries. The GDP for the US was 2.9% in 2018 (BEA, 2019) but more importantly there was a serious labour shortage in low wage sectors which migrant labour was in high demand (Dunkelberg et al. 2018) which is the small business report that recorded hiring difficulties and high levels of unfilled job openings. Sectors such as domestic work, construction, agriculture and cleaning services which many trafficking victims working in this industry have been connected to debt bondage (Polaris,2017). Many traffickers exploited this demand by promising jobs or smuggling workers into low regulated environments. Federal U.S policy shifted from the beginning of 2017 which moved to a tighter migration channels brought on by the Trump administration. The UNODC found that legal pathways constricted and homeland law enforcement on the U.S – Mexico border increased. The demand for using intermediary smugglers rose which gave increased discretion to traffickers to profit maximise in their best interestd (UNODC, 2018; Joarder and Miller, 2013).

Methodology

Data sources

The main dataset used for this thesis is the Global Report on Trafficking in Persons (GLOTIP). This dataset comes from national law enforcement agencies and criminal justice systems where the data is collected a process before submitting it to the UNODC. The UNODC provides indicators of classification of how a trafficked person is identified. Such as Detected trafficking victim, Persons suspected, prosecuted or convicted and trafficking offenses. Once this data is submitted, the UNODC reviews the data for internal consistency before adding it to the dataset. For this thesis the indicator “Detected trafficking victims” is the key variable of interest. While this dataset provides valuable insights into detection trends and judicial responses, it does not capture the true prevalence of trafficking, reflecting instead the reporting capacities and legal frameworks of the Member States. When look at how political instability in Honduras affects the rate of human trafficking to the U.S. The main forces of this will be placed on cross national data covering a panel of countries over the period 2005 to 2022. Allowing for both pre and post treatment analysis using the 2017 Honduran election crisis as the shock event. In order to comply with the SCM the data has annual panel of Honduras plus a pool of creditable donors made up of countries from Latin America. The donors were not hit by comparable domestic shock around 2016 – 2018. It is also important to have substantial pre-period trends in order to improve the fit and lower the bias as best as possible. The homicide estimate was also captured from the UNCDO.

The macroeconomic covariates are taken from the world banks World Development Indicators (WDI) database which was access through the World Bank API. This includes five variables, GDP per capita which is compiled from national accounts provided by countries to the UN and IMF and is expressed in US dollar. Unemployment rates are estimates from the International Labour Organization (ILO) based on national labour surveys. Net migration flows are estimated from the UN population division world population prospects (WPP). Urban populations are based on the UN’s World Urbanization Prospects which standardizes census definitions of “Urban”. Codes used to access API can be found in the appendix.

There are some limitations to the datasets such as underreporting of trafficking cases and a reliance of U.S based identification which may not fully capture victims. There is no individual level microdata available for Honduras, only country level aggregates. Despite these limitations, the data used is consistent with existing approaches in the literature (e.g., Akee et al. 2010; Cho et al. 2012) and allow for rigorous causal estimation using SCM and DiD frameworks. There is an argument to be made that due to the fact that there is not a very large number of pre treatment periods in this dataset, that the model is somewhat uncertain. Ferman et al. (2020) show in their paper that even with a large number

of pretreatment periods it is still possible to “hack” any analysis in order to find statistical significance that would be suitable for ones priors.

The model

Estimation of the synthetic Honduras

Let $i \in \{1, \dots, J + 1\}$ be the index of the countries and $t \in \{1, \dots, T\}$ is the set for years. The treated unit is Honduras which can be represented as $i = 1$ and the donor pool is $i = 2, \dots, J + 1$. The victim counts which is the observed outcome variable of choice for this paper is represented as Y_{it} . The unobserved synthetic which will be the counterfactual for Honduras will be represented as Y_{1t}^N . The pre treatment years which are 2005 – 2016 can simple be $t \leq 2016$ and the post treatment years are $t \geq 2017$. The actual treatment year is $T_0 = 2017$. The non-negative weights are sum up to one in order to satisfy $\sum_{j=2}^{J+1} w_j = 1$, where the weights are made up of the optimal combination of $W = (w_2, \dots, w_{j+1})'$. All pretreatment covariates and lagged outcomes for Honduras are represented as K and stored in the X_1 vector ($K \times 1$) and donors are stored in the X_0 vector ($K \times J$). The weights solve the aggregation minimisation function;

$$\hat{W} = \arg \min_{W \geq 0} (X_1 - X_0 W)' V (X_1 - X_0 W)$$

Where V is a positive semidefinite matrix that upweights predictors that improve the pre-fit. The synthetic counterfactual for Honduras is $\hat{Y}_{1t}^N = \sum_{j=2}^{J+1} \hat{w}_j Y_{jt}$ where $t = 1, \dots, T$. The gap between the actual Honduras and the synthetic Honduras can be defined as $\hat{\tau} = Y_{1t} - \hat{Y}_{1t}^N$, $t > T_0$. This follows abadie et al. (2010) and Cunningham (2021) protocols except where the potential outcome equation is defined as $Y_{1t} = Y_{1t}(0) + \alpha_{1t} D_{1t}$.

The convex hull assumption takes the treated units pretreatment characteristics and looks to see how they are represented as a convex combination of the donor pool. The actually convex hull is the weighted average of the donor pool. The value of testing this assumption is if it fails then the SCM will not be able to reproduce the treated units pretreatment trajectory well. If Honduras lies outside of the convex hull of all other countries in the donor pool, there will be no convex combination of donor units that will be able to form a trajectory before 2017 (pre-treatment). If this fails, then the SCM is not strong at building the counterfactual for what would have happened to the outcome variable had the treatment have not taken place.

Pre and post treatment fit are summarized by the mean squared prediction error (RMSPE):

$$RMSPE_{pre} = \sqrt{\frac{1}{T_0 - 1} \sum_{t \leq 2016} (Y_{1t} - \hat{Y}_{1t}^N)^2}$$

$$RMSPE_{post} = \sqrt{\frac{1}{T - T_0 + 1} \sum_{t \geq 2017} (Y_{1t} - \hat{Y}_{1t}^N)^2}$$

The RMSPE is used to calculate the goodness of fit in the pre and post treatment groups and a ratio between the two are calculated the quality of fit. Once that ratio is uncovered then a balance table is taken to check the quality of fit, lastly a predictor weights table is used to check the predictive power of the SCM.

The null hypothesis of no treatment effect is tested in two ways. The first being the size and persistence of the estimated treatment effect $\hat{\tau}$ are evaluated relative to the pre-treatment RMSPE. A large post-pre intervention RMSPE ratio for Honduras suggests that the estimated effect is unlikely to be explained by chance alone. This approach follows Abadie et al. (2010).

Due to the quasi-experimental design of the SCM the average treatment effect on the treated (ATT) is identified under the assumption that the weighted donor units provide a valid approximation of Honduras in the absence of the intervention.

The dependent variable will be the summation of the annual count of victims of human trafficking from Honduras into the U.S. This data is sourced from UNODC and the US Department of States Trafficking in Persons (TIP) reports. Where possible the data was supplemented by information from the International Organization for Migration (IOM) and the US department of Homeland Security (DHS). The TIP report outlines the minimum standards for the US Trafficking Victims Protection Act which ranks based on three tiers. Tier 1, Tier 2 and Tier 3 with Honduras consistently been placed in Tier 2 or Tier 3 Watchlist. The result of this means that Honduras is consistently rated as having institutional weaknesses in combating trafficking. The U.S is a Tier 1 country, and the summation of the reports comes from the destination country enforcement records and since trafficking is a federal crime in the U.S, the cases that are reported are done so in a standardized way even if there is under reporting present.

The treatment event is the 2017 Honduran presidential election crisis which resulted in mass protest and crackdowns for the government as a result. The shock is treated as an exogenous event that plausible affects migration patterns for vulnerable people in to be trafficked. The treatment will be

encoded using binary variable. That being 2017 and onwards is the treatment group. The outcome variable which will be use is victim count which is define as the annual total number of people who were classed as having been trafficked by the U.S authorise. Since it is a federal crime in the U.S to traffic an individual as well as the U.S being classed as a tier one country in relation to classification of trafficking. Since there was a supreme court appeal form the Honduras president in ~2014 about being able to re-run for another election. There needs to be made certain that there are no anticipation and the shock in 2017 was an idiosyncratic shock. Considering that before 2017 there was no outright protests like those seen in 2017 except for coup of 2009 but that unrest was not at the same level of 2017.

The donor countries

The donor countries need to have no interference from the 2017 Honduras event.

The donor pool consists of El Salvador, Guatemala, Nicaragua, the Dominican Republic, Colombia, Ecuador, and Paraguay. These countries were chosen because they share important structural similarities with each other that can be captured in the data in order to build a solid donor pool for Honduras that matches and is relevant to human trafficking dynamics. All these countries capture in the donor pool are recognised by the UNODC as source countries for trafficking victims. The predictors used in the synthetic control model such as GPD per capita, unemployment, net migration, urbanisation, population size and homicide rate.

Geographic and socio-political comparability is especially important for capturing a common regional shock. For example, US change in immigration enforcement due to administration change. Which can shift the migration corridors and fluctuations in remittance flows which will affect the entire Northern Triangle and those neighbouring states. At the same time none of the selected countries in the donor pool should experience a shock of the same magnitude of that seen in Honduras 2017 post-election unrest. This strengthens the credibility of the synthetic control since any divergence after 2017 can be attributed purely to Honduras domestics treatment rather than to broader factors.

There is an assumption of no interference from the donor pool on the outcome affected by the intervention on the treated unit. An example of when there might be spillover would be an event like the Arab Spring that started in Tunisia and spread to Egypt, Libya and Syria. In the case of Honduras there was no contagion effect between neighbouring countries.

Countries in the donor pool include:

- El Salvador
- Guatemala
- Nicaragua
- Dominican Republic
- Colombia
- Ecuador
- Paraguay

When designing a donor pool country with policies/events even close to the 2017 election crises when building a relevant counterfactual. More specifically checking for similar shocks in the dataset, Abadie (2020) states that units in the donor pool should similarly events. El Salvador has had no Honduras counting crisis and national elections were broadly assessed as credible with no major tabulation held or OAS call to re check any votes. Guatemala featured serious judicial interference attempts after the vote count, but this was in 2023 which is outside of the window for this study. Nicaragua has elections in Nov 2016 which were widely criticized for being systemic bias and followed by this, in 2018 there was massive protests which were met with lethal repression. The credibility of election is therefore a problem, and the country should be dropped from the donor pool. The Dominican Republic had multiple elections being suspended mid-day which was considered disruptive but there were not mass protests and unrest in the country. Colombia has no comparable national level crisis's except for some isolated cases of fraud allegations but observer missions reported overall functionality and no late count reversals. Ecuador has the closes analogue within the donor pool which say in 2017 president election ended very close and alleged fraud. There were no wide scale protests like those seen in Honduras. Paraguay saw a serious political crisis in 2017 which saw mass protests and congress was set on fire over a constitutional amendment to allow presidential re election were one protester was killed. This was not a vote count controversy like Honduras but a democratic rule crisis. It should be dropped however due to it being similar to Honduras.

Potentially “treated like” shocks to avoid in the donor pool are Paraguay and Ecuador because of shocks caused by election fraud around the same time. This is supported in Peri and Yassenov (2019) paper by displaying a discipline donor selection to avoid structurally dissimilar control to that of the treatment event.

(Cunningham, 2021, SCM)

Building a synthetic control in R

The R package that was used to build the synthetic control is called Synth developed by Abadie et al. 2010. The package was built to estimate the causal effect of policy interventions and other events of interest and is well suited to investigate events at the aggregate level. This method constructs a convex combination of all the donor countries that best reproduces Honduras pretreatment predictors and those of the outcome path. The donor weights W must be sum to one and be non-negative and all the predictor weights V are chosen to best minimize the pretreatment MSPE so that the synthetic outcome $Y_{1t}^* = \sum_j w_j Y_{jt}$ approximates Honduras prior to 2017. The donor pool is restricted to Latin America and uncertainty is quantified via a placebo in space randomization inference with a pre-fit filter of K .

The covariates chosen to construct a credible SCM for Honduras so that the counterfactual is justifiable are as follows, GDP per capita, Homicide rate, Political stability rate, Unemployment rate and net migration rate. These covariates are widely recognised in migration and trafficking literature and deemed credible. GDP per capita will capture the economics health of the economy. This will serve as both a push and pull factor. A “push” factor in relation to when the economy is doing poorly and people become desperate and on the flip side of this, the “pull” factor of the strong US economy provides an incentive for better standard of living (Case and Deaton, 2015). The unemployment rate is a further reflection on the labour market pressures; this is included because it captures market pressures that are not inclusively included in GDP alone. A persistent level of elevated unemployment has been shown to increase migration incentives between countries. The homicide rates are a good proxy for security of the country, high levels of violence have been identified as strong drivers of displacement and trafficking in central America and especially relevant in the Honduras context (Hiskey et al. 2018; Lang, 2013). The political stability index covariate captures how legitimate the government officials are. The U.S might have elevated levels of homicide in certain states; however, the institutes are classified on their legitimacy. In the case of Honduras, the legitimacy of government deteriorated sharply during the 2017 election crisis. Political stability has been lined in the wider literature to institutional shocks (Aisen et al. 2013). Lastly the net migration rate offers a demographic proxy for the outward movement pressures of people. This aligns the donor pool with Honduras. Collectively the choice of these covariates covers the economics, security and institutional mechanisms most relevant to trafficking flows.

Covariates used for matching include:

1. GDP per capita (World Bank, WDI)
2. Homicide rate (UNODC)
3. Political stability index (World Governance Indicators)
4. Unemployment rate (WDI)
5. Net migration rate (UN Population Division)

Placebo and Robustness Tests

The use of a placebo in space design has been implemented on the Honduras post treatment divergence. This allows for each donor to be turned into a “pseudo-treated” unit and the SCM was re-estimated with the remain donors as controls. The only placebos that are retained are those with an acceptable pre-fit to try and minimize any extreme values from poorly matched units. Pre/post RMSPE ratios for Honduras and for all remaining placebos have been kept and the exact p-values have been reported. This procedure has been taken from the guidelines of Abadie et al. (2010)

$$pretreatment\ RMSPE \leq K * RMSPE_{HND}$$

Pointwise permutation test

Due to the large ratio being possible to be caused by a short-lived spike, this paper complements the ratio test with a year-by-year pointwise permutation p-value. The standardized each units gap by its own per-period RMSPE and the post-2017 year. The benefit of this is that the placebos which has a standardised gap that is larger than Honduras are only kept and those lower are removed. This mirrors the applied migration SCM work by Peri & Yassenov in the Mariel Boatlift study which pairs global divergence with year-specific placebos comparisons in order to assess both the timing and persistence (Peri et al. 2019)

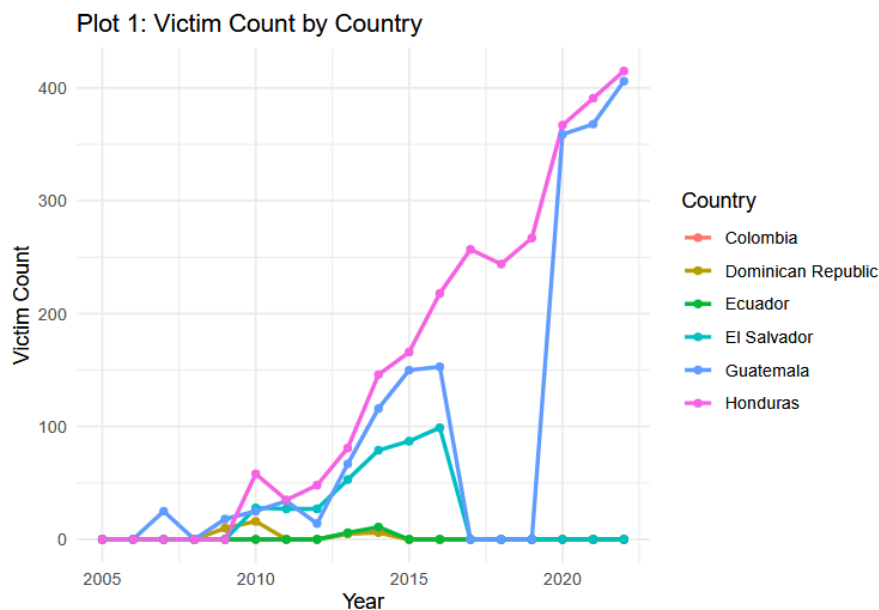
Donor data sensitivity: backfilling the clean donors specification.

Several donors exhibit a zero or missing outcomes for the years 2017-2019 due to potential reporting discontinuities. In order to test that these potential missing values did not inflate Honduras post 2017 gap, a two-step sensitivity protocol was carried out. The first step was backfilling the missing values for the donors only from 2017-2019 using donor specific linear pre-trends. Honduras and all pre-treatment outcomes remain untouched. The second step re-estimate the clean donors and drop any outcomes that are zero/NA and re-estimate. After this the average treatment effect (ATT) by year was re-estimated.

Leave-one-out over donors

Due to baseline weights have a single donor for the SCM, the use of leave one out stress test was used to re-estimate the SCM but exclude Guatemala (the single donor weight). The resulting pre/post RMSPE, ATT and gap plot were obtained. Following standard SCM practice (Abadie et al. 2010; Peri & Yassenov, 2019; Cunningham, 2021), conducting leave-one-out donor test for high weight donors in the SCM are useful in assessing the sensitivity of donor concentrations.

Secondly, permutation testing is employed by iteratively reassigning the treatment to each country in the donor pool. For each placebo country, a SCM is constructed, and placebo treatment effects are estimated. From this the distribution of “placebo effects” under the null of no intervention can be obtained. The exact p-value is found by the proportion of placebo units with an estimated post/pre intervention RMSPE ratio greater than or equal to Honduras observed ratio.



Plot 1 shows the victim rate of identified human trafficking victims from Honduras and other selected donor countries in Latin America between 2005 and 2022. There are two patterns that can be visually seen in this plot, the first is from 2010 onwards there is an increasing amount of victim year after year. This is likely due to the coup experience in Honduras in 2009, there is however not enough data before the coup occurred to build a creditable experiment like SCM. The actual model would only have four years of pre – treatment data to work with. This trafficking trend does keep increasing since 2010 to over 400 victims in the U.S by 2022, far higher than any other country in Latin America that has reported values. This suggests high “push” incentives for Honduran citizen to try and get the U.S.

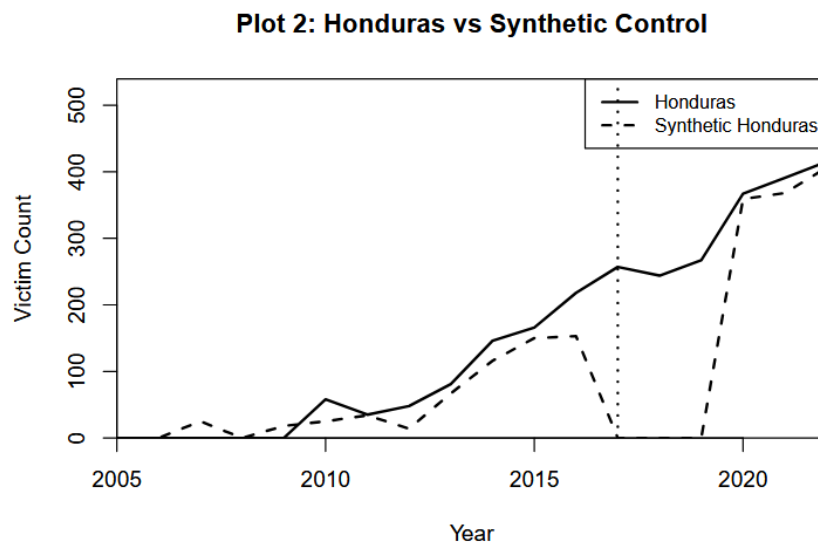
In 2017 there is a very sharp discontinuity in reporting, leading to an abrupt drop to zero trafficking victims from all countries except Honduras. This is not plausible as a real-world reflection of Latin America trafficking dynamics, rather it is a data reporting issue. Potentially due to the first Donald Trump Administration. There is no official record of the Trump administration halting data collection, it is likely that slashed fundings for anti-trafficking efforts and tougher immigration enforcement contributed to the sharp fall in reporting. The anomaly shows a further limitation of the dataset. For this reason, the SCM relies on valid pretreatment and donor countries for comparisons where the

“artificial” zeros could bias estimators if not handled properly. Because of this the empirical strategy restricts attention to pre 2017 data for construction of the SCM and interprets post 2017 patterns with caution.

There is evidence to suggest good heterogeneity across countries pre 2017 treatment due to Honduras showing enough variation before the intervention in order to build a counterfactual the donor anomalies are smoothed by weighting.

Results section

Baseline results



Human trafficking rate in Honduras vs synthetic Honduras:

Plot 2 compares the trajectory of Honduras against that of the Synthetic Honduras, it shows the annual victim count of those individuals being classified as trafficked in the U.S who are of Honduran citizenship. The solid line is the actual rate and the dashed line in the counterfactual over the period 2005 – 2022. The vertical dashed line in the treatment year of 2017 which was the Honduran presidential election crisis. Avdan et al. 2023 suggests that entry points, modalities or jurisdictions can violate the stable measurement environment and dilute post treatment gaps which is what is observed above.

Prior to the intervention the synthetic Honduras tracks the actual Honduras quite closely from the years 2005 - 2016, with good parallel trends. This goodness of fit for the SCM shows that the donor pool was successfully able to reproduce the underlying trajectory of Honduras before the treatment. After 2016 there is a sharp decline for the Synthetic Honduras tends towards zero and after 2017 the victim counts are near zero, this suggests large scale reporting collapse in the donor countries, and the synthetic control recovers in 2019. In contrast to this, Honduras remains elevated with the victim counts continuing to trend upwards year after year which stabilised around 250 – 400 victims annually. The true treatment effect, Honduras sustained a rise in victims while its counterfactual falls to near zero. This suggest that the 2017 political crisis exacerbated trafficking vulnerabilities in ways not seen in other countries in the donor pool. The post 2017 gap supports the hypothesis that political

instability acts as a strong push factor in human trafficking, even though the donor pool has anomalies, there is seen to be persistence of elevated victim counts relative to the donor pool.

The findings of the above plot comparing Honduras and the synthetic Honduras show a diverging path. This however did start before 2017 which is the treatment year of interest. Therefore, based on the visual inspection of the graph, the 2017 presidential crisis cannot be causal linked to the event. The cutoff for this donor pool is 2022 since the UN only release datasets every two years. The rates did increase year on year for Honduras even during the pandemic. It was surprising to find counts during the Covid-19 pandemic were still elevated. One would think that the drop to zero would have occurred during the pandemic and not before it.

Table 1: Pre-/Post-treatment RMSPE and Ratio for Synthetic Control

Treated_Unit	Treat_Year	Pre_RMSPE	Post_RMSPE	RMSPE_Ratio	N_Pre	N_Post
Honduras	2017	27.037	181.452	6.711	12	6

Table 1 shows the trajectory of the reported trafficking victims counts for Honduras and the Synthetic control for Honduras. In the years before the treatment event, prior to 2017 the pre treatment is a strong fit with the SCM tracking that of Honduras closely. Due to this, there is support of the validity of the donor pool and the predictor variables in being able to produce a counterfactual that is valid. After the election crisis a divergence between Honduras and its synthetic starts to show. Honduras has an increasing victim count than the synthetic which could suggests a possible increase in trafficking flows due to the election crisis. However, this divergence is not strong and is not pronounced compared to placebo in space tests below. The treatment effect may be artificially inflated due to many of the donor countries values dropping to zero after the treatment when Honduras continued to be increasing in count year after year. Therefore, even though the treatment effect is consistent with the hypothesis that the election crisis affected trafficking outcomes, the evidence is not robust and any interpretation from this should be view with caution.

Balance table (standardized approach due to mix of signal)

Table 2: Predictor balance (pre-treatment): Honduras vs Synthetic control

Predictor	Honduras	Synthetic	DonorMean	Gap
gdp_pc	1896.210	3037.744	4543.009	-1141.534
unemp	4.674	2.931	5.678	1.743
urban_pop	52.209	48.573	65.839	3.635
net_mig	-5788.500	-39605.000	-34890.767	33816.500
population	8449674.417	14662446.374	18147322.083	-6212771.958
homicide	3829.167	4724.083	5749.615	-894.917
estimate	-0.829	-0.685	-0.578	-0.144
special.victim_count.2007	0.000	25.000	5.000	-25.000
special.victim_count.2009	0.000	18.000	5.600	-18.000
special.victim_count.2010	58.000	25.000	13.800	33.000
special.victim_count.2011	35.000	34.000	12.200	1.000
special.victim_count.2012	48.000	14.000	8.200	34.000
special.victim_count.2013	81.000	67.000	27.200	14.000
special.victim_count.2014	146.000	116.000	43.800	30.000
special.victim_count.2015	166.000	150.000	47.400	16.000
special.victim_count.2016	218.000	153.000	50.400	65.000

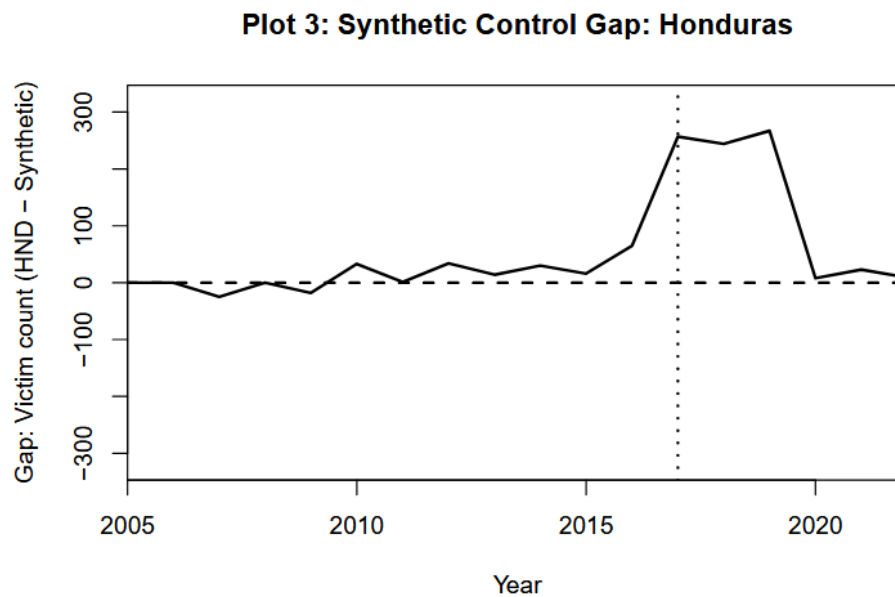
The balance table seen in table 2 shows that the outcome lags are reasonable case for the years ranging 2011 – 2015 and 2016 is off by around 65 on a base of 218 which is ~30%. This is the main pre-fit miss in the SCM. For this reason, it makes more sense to report on the standardise balance and predictor rates since the balance table is a mix of counts and rates which the optimizer struggle to interpret.

There are large imbalances observed in this table 2 , which can be viewed in population size where Honduras is notable smaller than that of the synthetic (~ 8 vs 15 million). There is also a problem in the net outflow covariate where Honduras experiences large outflow versus that of the synthetic control. The gaps in the balance table suggest that the donor pool was not fully successful of reproducing Honduras demographic dynamics.

The pre and post RMSPE were calculated using the below equation, this shows the typical size of the Honduras vs synthetic Honduras gap. The pre gap is around 27 units and the post gap is around 181 (table1). The pre gap means that the SCM performed fairly well missing around 27 victims per year whereas after the treatment the gap jumps to 182 victims representing the estimated treatment effect. The ratio between the two is ~ 6.71, meaning that the deviation is 6.7x larger than that seen the pre period noise.

$$\frac{RMSPE_{post}}{RMSPE_{pre}} \approx 6.71$$

Gap in trafficking rate for estimation before and after treatment. Plot 3 of Synthetic Control Gap: Honduras plots the difference over time. Prior to 2017 the gap is around zero, which provides evidence that the synthetic unit tracks Honduras closely in the pre treatment period. After the intervention in 2017 that gap increases sharply where it is seen to peak at an estimated 250 victims above the synthetic estimates between 2018 and 2019. Again, suggesting that Honduras experienced exceptionally higher levels of trafficking rates relative to that of the counterfactual immediately after the intervention. Then after 2020 that gap narrows again back to zero. The reversal provides further evidence that the intervention event was short lived shock and diminished after three years.



Donor table

Table 3 shows that the synthetic Honduras is composed entirely of Guatemala which receives a weight of 1, whereas all the other donor countries El Salvador, Ecuador, Dominican Republic and Colombia have a value of zero. Peri and Yassenov (2019) accept that weight vector contains a small number of donors and using a single donor is not ideal but acceptable. This suggests that Guatemala

Table 3: Synthetic control weights (donors with positive weight)

	Donor country	Weight
5	Guatemala	1
4	El Salvador	0
3	Ecuador	0
2	Dominican Republic	0
1	Colombia	0

alone was sufficient enough to build the synthetic control for Honduras in the pre treatment assignment under the optimization procedure. While this is statistical valid, the synthetic is not very diverse and makes the counterfactual dependent on only the trajectory of Guatemala. This is likely due to other donor countries have a level of zero victim count after 2017 and Guatemala is the only country in the donor pool to return to similar levels after 2019. The consequence of this is that robustness is limited especially if there are unobserved idiosyncratic shocks in Guatemala over the same study period. Backfilling is used to address this robustness issue in table 8 but these units are only useful for gaining a better understanding of sensitivity and are going to be bias.

Table 4: Synthetic control weights

	Predictor	Weight
1	gdp_pc	0.122
16	special.victim_count.2016	0.095
4	net_mig	0.094
10	special.victim_count.2010	0.088
7	estimate	0.086
14	special.victim_count.2014	0.078
12	special.victim_count.2012	0.076
13	special.victim_count.2013	0.070
3	urban_pop	0.062
5	population	0.061
6	homicide	0.058
15	special.victim_count.2015	0.053
11	special.victim_count.2011	0.042
2	unemp	0.013
8	special.victim_count.2007	0.002
9	special.victim_count.2009	0.001

Table 4 outlines the weights which are assigned to the predictors in the optimization. The resulting table produced the order of importance for each covariate. GDP per capita (0.122), net migration (0.094) and political freedom index “estimate” (0.086) have the greatest importance. As well, several lagged outcome variables are of importance, 2016 (0.095), 2010 (0.088), 2014 (0.078), 2012 (0.076), and 2013 (0.070). The predictors with the lowest weights are unemployment and earlier victim counts which all received negligible weight which suggests they have little impact on improving the quality of the fit for the model. The main issue here is that the synthetic is based purely on one donor, this however is not uncommon in the literature as seen in (Abadie et al. 2010) where single-country synthetics controls are accepted as valid counterfactual. The standardized table in table 4 shows a V-matrix of predictor weights generated from the synth process. The model places around 50% of total weight on lagged outcomes which is the sum of all victims counts, therefore the SCM is using the pre-treatment victim count trajectories in order to anchor the counterfactual. This entire vector of weights sums to one as is a requirement for SCM. The outcome variable itself (special.victim.count) did achieve a relatively close fit for most pretreatment years, this divergence in the years from 2016 onwards where Honduras records a far larger victim counts. The synthetic therefore underestimates the rising pretreatment trend which could also contribute to a bias post treatment divergence.

Additionally, this under fit helps to explain the high pretreatment RMSPE in table one as well as the high post/pre ratio of 6.71. The major issue is that the synthetic does not closely match Honduras sharp upwards trend in the years immediately before 2017.

ATT table

Table 5: Post-treatment effects by year

	Year	ATT (Honduras – Synthetic)
2017	2017	257
2018	2018	244
2019	2019	267
2020	2020	8
2021	2021	23
2022	2022	9

Table 5 displays the post treatment effects for the average treatment effect on the treated unit. It is calculating the difference between the actual Honduras and the synthetic. All results are positive with substantial values in the years immediately after the election crisis with an estimated 257, 244 and 267 more victims in the years 2017, 2018 and 2019. These large differences are consistent with the Honduras synthetic plot: Gap. These effects diminish sharply from 2020 onwards, the reversion towards zero suggests that the divergence between the synthetic and the actual Honduras was only temporary. One suggestion is that the ATT shows that the 2017 election crisis had a large effect on trafficking rates, but a more plausible explanation is that the sharp decline reflects disruptions in reporting across the donor pool. It is still worth formally assessing for statistical significance of these differences.

Table 6 reports a the RMSPE ratio of 6.711 which suggest substantial collapse of goodness of fit after the intervention. When benchmarked against the placebo countries Honduras does not emerge as an outlier, with an exact p-value of 0.40. Which means that 40% of placebo units display ratios at least as large or larger. Therefore, across the entirety of the post treatment periods, Honduras cannot be distinguished from chance variation and causality cannot be shown. This result is expected based on donor heavy control weights. In table 7 a pointwise permutation p-values are taken in order to see the full breakdown of placebos.

Table 6: Randomization inference summary (ratio-based exact p)

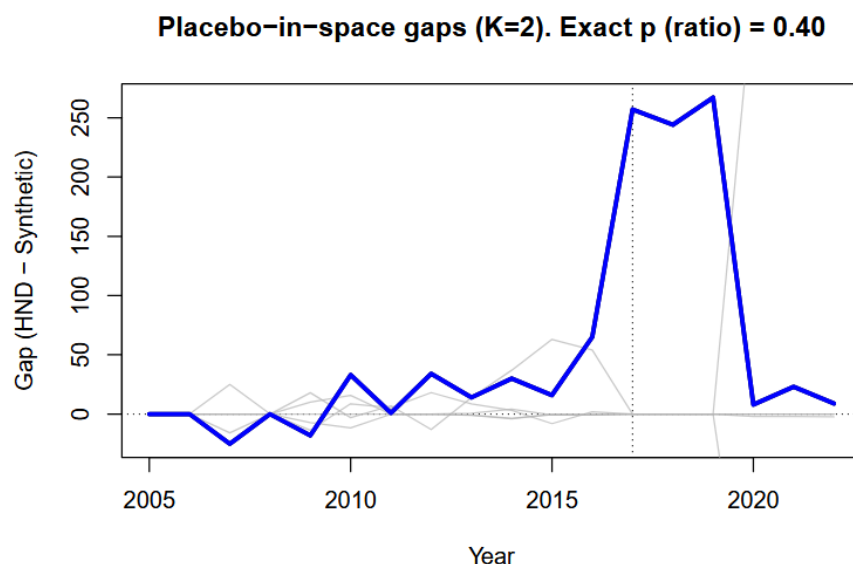
Pre_RMSPE	Post_RMSPE	RMSPE_Ratio	Placebos_Kept	Exact_p_Ratio
27.037	181.452	6.711	5	0.4

Table 7 shows that the gaps in Honduras between 2017-2019 are highly unusual with p-values of 0.0 initially and from 2020 onwards the p-values rise to 0.2 which is consistent with earlier findings that Honduras does not stand out from the donor pool after the initially divergence fades. This suggest that the ATT estimates are bias which is due to a single weight being used in the donor pool.

Table 7: Pointwise permutation p-values (standardized gaps)

Year	Pointwise_p
2017	0.0
2018	0.0
2019	0.0
2020	0.2
2021	0.2
2022	0.2

The below plot visualises the placebo-in-space results. With the blue line representing Honduras and the grey lines represent donor countries which were re-assigned as placebos. Again, even though Honduras presents the larges post-treatment divergences the plot shows many placebos that display the gaps. This result is consistent with the p-value of 0.40 which further reinforces the conclusion that Honduras is not a valid outlier across any of the other countries in the donor pool.



Randomization inference following that outlined in Cunningham (Cunningham, 2021) procedure yields a large post 2017 increase in the reported trafficking rates relative to the Honduras counterfactual. After conducting the placebo-based randomization inference it is found that it is limited to a small valid placebo pool which is four donors at $K = 2$. The overall exact pointwise P-value of 0.2 for each year based on the RMSPE. The mean standardized post gap tests yield $p = 0.20$.

These results imply insufficient statistical evidence that the 2017 election crisis in Honduras can be causal linked to an increase in trafficking rates.

Honduras exhibits a front-loaded increase trajectory in victim counts through the years (2017 -2019) but statistically this is not distinguishable from a given chance due to the small valid placebo pool and issues with predictors. Therefore, this SCM and testing does not support the original hypothesis that the 2017 election shock in Honduras increased trafficking rates to the U.S in a causal sense when looks at the time frame allocated for this thesis.

Limitations

There are three major issues with this model and the results of it. Firstly, there is low placebo power where only four donors pass the pre-fit screen. Secondly, there is a scale mismatches that are likely inflating the goodness of fit errors. Lastly, the outcome variable of interest which is victim counts which is a reflation of reporting intensity and not a true value. This has led to results that are null and inconclusive. Using the SCM method outlined by Abadie et al. (2015) and Cunningham (2021) there is no statistically significant causal effect.

Robustness check: Backfilling Donor Data

Table 8: Backfilled donor victim counts (2017–2019)

Country	Year	Victim Count
Colombia	2017	3
Colombia	2018	3
Colombia	2019	3
Dominican Republic	2017	4
Dominican Republic	2018	4
Dominican Republic	2019	4
Ecuador	2017	4
Ecuador	2018	4
Ecuador	2019	5
El Salvador	2017	96
El Salvador	2018	106
El Salvador	2019	116
Guatemala	2017	141
Guatemala	2018	155
Guatemala	2019	169

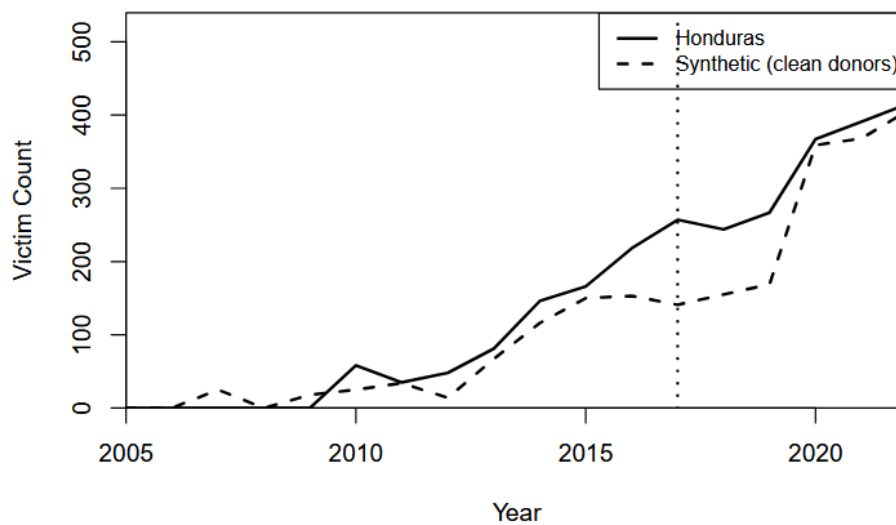
For a robustness exercise, the synthetic control was re-estimated after backfilling missing donor outcomes for 2017–2019. Linear trend extrapolation was used to fill reported zeros in these years despite having positive pre-treatment victim counts. Table 8 shows the results of the countries that were backfilled successfully with 15 values included. This removed donors that recorded a zero/NA outcomes from 2017 -2019.

Even with the adjustment the results remain effectively unchanged. The optimization procedure continued to assign all the donor weight to Guatemala with other donors receiving zero weight. This outcome shows that even after imputation of the backfilled data, Guatemala remained the closest match to Honduras in terms of pretreatment outcomes and the covariates. The lack of change demonstrates that not all of the variation in the donor pool can be put down to missing data alone. Instead, this robustness check suggests that selection of the country of Guatemala as the sole donor is driven by strong similarities between that of Honduras and potentially not so much the reporting gaps.

Table 9: Synthetic control weights (donors with positive weight)

	Donor country	Weight
5	Guatemala	1
4	El Salvador	0
3	Ecuador	0
2	Dominican Republic	0
1	Colombia	0

Plot 5: Honduras vs backfilled synthetic



To address sparse post-treatment missing values in control units, we implement a donor-only backfilling procedure for 2017–2019, leaving the treated unit and all pre-treatment observations unchanged. Conceptually, this follows the interactive fixed effect view of panel outcomes used in economic policy evaluation (Gobillon & Magnac, 2016, *RESTAT*). This above sensitivity plot shows a good pretreatment fit and a somewhat good front-loaded treatment effect with a noticeable gap in the year 2017 onwards where the synthetic collapsed to zero. Even this specification pulls the synthetic up and weakens the case for a large effect and therefore leads the conclusion to fail to reject the hypothesis of a sizeable and statistically significant causal impact. The dashed synthetic closely tracks Honduras from 2005 – 2016 which supports the credibility of the counterfactual design. In 2017 the visual level difference is about 90 victims, this is a large one time jump not a collapse. From 2018–2019 there is a suggestion of a temporary shocks rather than a lasting regime shift due to the election crisis. From the years 2020 – 2022 the lines converge to have little to no sustained effect. The earlier large effect was primarily due to mechanical design flaw within the data where the donor went to zero in 2017-2019. Using backfilling from the donors pre trends and dropping positive weight donors with zero post year eliminated that artifact. The remaining pattern is economically meaningful but statistically fragile. This is formal tested below using randomization p-values.

Table 12: Predictor balance (pre-treatment): Honduras vs Synthetic control

Predictor	Honduras	Synthetic	DonorMean	Gap
gdp_pc	1896.210	3037.744	4543.009	-1141.534
unemp	4.674	2.931	5.678	1.743
urban_pop	52.209	48.573	65.839	3.635
net_mig	-5788.500	-39605.000	-34890.767	33816.500
population	8449674.417	14662446.374	18147322.083	-6212771.958
homicide	3829.167	4724.083	5749.615	-894.917
estimate	-0.829	-0.685	-0.578	-0.144
special.victim_count.2007	0.000	25.000	5.000	-25.000
special.victim_count.2009	0.000	18.000	5.600	-18.000
special.victim_count.2010	58.000	25.000	13.800	33.000
special.victim_count.2011	35.000	34.000	12.200	1.000
special.victim_count.2012	48.000	14.000	8.200	34.000
special.victim_count.2013	81.000	67.000	27.200	14.000
special.victim_count.2014	146.000	116.000	43.800	30.000
special.victim_count.2015	166.000	150.000	47.400	16.000
special.victim_count.2016	218.000	153.000	50.400	65.000

Table 12 shows the predictor balance table for the backfilled donor pool countries with missing values after 2017; the table should be the exact same as the original baseline balance table. This is because nothing else has changed in the data structure before 2017. The table is identical, so nothing has been changed before 2017 and only donor pool countries with missing values have been altered.

Cumulative post gap for backfilled donors

Plot 6 shows a cleaned donor pool with a front loaded temporary positive effect that Honduras synthetic gap that jumps sharply at 2017 (≈ 120 victims). And remains elevated through 2018-2019 before shrinking back to ≈ 20 victims through 2020-2022. The pre treatment gaps oscillate by roughly 30 victims which is not perfect but acceptable for the pre fit. Overall, the effect is short lived and moderate which is same trend seen in the baseline SCM.

Plot 6: Gap for backfilled donors



RMSPE for backfilled

Table 13: Summary of post-treatment fit, RMSPE ratio, and randomization inference

Pre_RMSPE	Post_RMSPE	RMSPE_Ratio
27.037	72.635	2.687

The pre treatment RMSPE is 27.04 which is the same as the original estimate, this supports the same pre treatment fit as before. The post RMSPE falls to 72.64 (down from 181.45) which yields a post/pre ratio of 2.69. This model still shows evidence of a deterioration in the fit after the 2017 intervention which is consistent with a post treatment divergence. However in this case the actual magnitude is smaller than the baseline. It is worth noting that a randomization-inference based on this ratio is smaller than the baseline synthetic which reinforces the cautious interpretation.

Table 14 shows the post treatment effects by year for the backfilled data. This ATT estimates that the effect is immediately felt directly after the election crisis in 2017. With results of +116 (2017), +89 (2018), and +98 (2019) difference between Honduras and the synthetic. After 2020 the difference in the ATT reverts back to a smaller value much like the first model. This pattern does align with the gap plot for the clean donors which shows a discrete post 2017 spike followed by a convergence.

The backfilled donors specification aligns with the initial results of a short run increase in reported victim counts post 2017 treatment, albeit with a smaller magnitude which is captured in the ratio. This implies that part of the larger baseline spike was inflated by donor countries with implausible zero counts post 2017. The backfilled donors provide suggestive but not conclusive evidence of a short lived shock caused by the election crisis that trafficking victims count did increase. The backfilling of donors from 2017 – 2019 suggest dropping the problematic donors reduces the peak gaps but this leaves the conclusions of the baseline synthetic unchanged. Guatemala still remains the best match even after backfilling.

ATT for backfilled

Table 14: Post-treatment effects by year

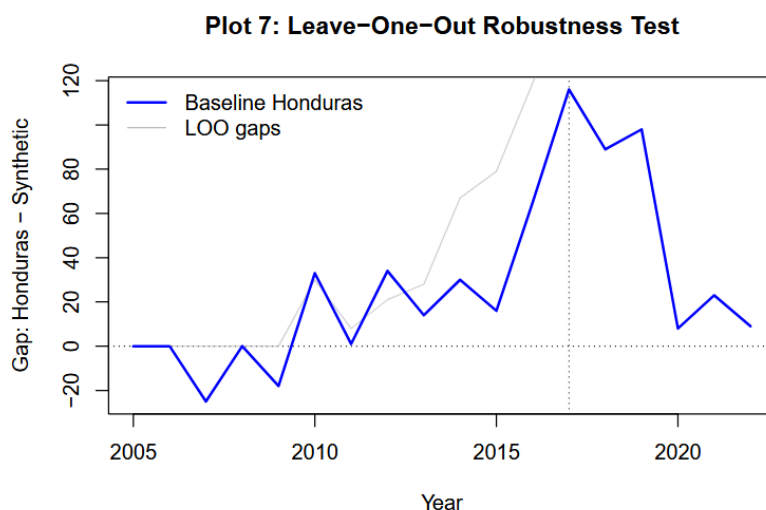
	Year	ATT (Honduras – Synthetic)
2017	2017	116
2018	2018	89
2019	2019	98
2020	2020	8
2021	2021	23
2022	2022	9

Robustness test: Leave-one-out

Table 15: Leave-one-out RMSPE diagnostics: effect of dropping each positive-weight donor

	Dropped	Pre_RMSPE	Post_RMSPE	Ratio
pre	None (baseline)	27.037	72.635	2.687
pre5	Colombia	27.037	72.635	2.687
pre1	Dominican Republic	27.037	72.635	2.687
pre2	Ecuador	27.037	72.635	2.687
pre3	El Salvador	27.037	72.635	2.687
pre4	Guatemala	47.504	296.525	6.242

To evaluate if the estimated effect is really driven by a single donor, the implementation of leave-one-out (LOO) test was performed and the synthetic control was re estimated for each non-negative donor. Table 15 show the reports of the RMSPE diagnostics and exclude Colombia, Dominican Republic, Ecuador and El Salvado which produced no change in either the pre or post RMSPE. This provides evidence that these donors carry no weight in the baseline specification. In contrast to this removing Guatemala was the only donor to be assigned a substantial positive weight. The pre treatment RMSPE rose from 27 to 48 and post treatment rose from 73 to 297 with a RMSPE ratio increasing from 2.687 to 6.242. These results show that Guatemala is the only anchor donor for building the Honduras counterfactual. This result is however suggestive only.



Looking at plot 7, the key takeaway is that Guatemala is the only synthetic control that appears to be fairly stable overall. By excluding all the other donors and the output produced was no meaningful shift in the RMSPE suggest that the donor pool is not sensitive to small changes at all. Therefore, the single donor used in baseline and backfilled procedure still holds as the only credible estimator for Honduras's synthetic. This is seen in the sharp increase in fit errors once Guatemala is removed from the possible donor selection.

Conclusion

The goal of this thesis was to use quantitative techniques to uncover if there is a statistically significant link between the 2017 Honduran presidential election crisis and human trafficking flows into the United States. Using the synthetic control method, a counterfactual was constructed for Honduras based on a donor pool of comparable Latin American countries. There is some evidence to view small short run effects but even these remain uncertain and lack robust support. A series of robust and inference test were carried out, such as the implementation of RMSPE ratios and placebo-in-space. Backfilling of reporting gaps using linear trend extrapolation was used to address the issue of a donor pool drop to zero after 2017. The finding for both these tests suggested that Guatemala was the sole donor weight. In order to give more evidence to this claim, leave-one-out robustness test was performed. The results indicate a short lived divergence between 2017-2019, that being after the treatment year there was an increase in traffic victim counts in the U.S. This effect reverted back to similar levels from 2020 onwards and these results were not statistically significant in randomisation inference conducted through placebo-in-place and leave-one-out testing.

The estimated average treatment effect is relatively small in comparison to the pre-treatment victim count variation rate which is suggestive that the 2017 Honduran election crisis did not have a persistent effect which caused the increase in trafficking flows into the U.S. The results of this thesis are inconclusive, but it does demonstrate an applied contribution at finding a causal link in a under researched but increasing socially problematic area of human trafficking. This thesis does also highlight how measurement in international reporting can have an effect on research methodology even though policy shifts like those seen in the U.S. in recent years. In order to advice policy makers on evidence based anti-trafficking policy proposals, it is important that the empirical tools that researchers have at their disposal are equally matched with reliable measurements. Even though the arguments based on common theory laid out by other researchers suggest there is an effect. It is down to limitations in the collection of information that makes it difficult to estimate the impact of trafficking dynamics.

Fragile trafficking conclusions are due to issues with reporting discontinuities and without better cross-country measurements large persistent effects can be ruled out in these data sets. The results of this thesis are non-conclusive and there is no evidence that the 2017 Honduras election crises had a causal effect on the increasing trafficking flows of victims into the United States. While the analysis does not validate the initial hypothesis it does add value by demonstrating how fragile trafficking evidence in the face of missing data. With improved datasets in the future researchers will be better equipped to answer important society questions and minimise the power of causality. This illicit market is growing year on year and without transparency in victim data collection, those victims trapped in the industry will continue to increase each year.

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Appendix

Variable	Indicator Code (World Bank API)	Definition / Construction	Unit	Primary Source
GDP per capita	NY.GDP.PCAP.CD	Gross Domestic Product divided by midyear population, expressed in current US dollars. Derived from national accounts data.	Current US\$	World Bank WDI; national statistical offices; IMF & UN Statistics Division.
Unemployment	SL.UEM.TOTL.ZS	Share of the total labor force that is without work but available and seeking employment. Harmonized to ILO definitions.	% of labor force	ILOSTAT (International Labour Organization), compiled in World Bank WDI.
Net migration	SM.POP.NETM	The net total of migrants (immigrants minus emigrants) during a five-year period, derived from population balances.	Number of persons	United Nations Population Division, World Population Prospects (WPP).
Urban population	SP.URB.TOTL.IN.ZS	People living in urban areas as defined by national statistical offices, standardized for comparability.	% of total population	UN Population Division, World Urbanization Prospects (WUP).
Population	SP.POP.TOTL	Total midyear population, based on births, deaths, and net migration.	Number of persons	World Bank staff estimates based on UN Population Division, World Population Prospects (WPP).